

Healthcare Multimodal Foundation Model System

Image + clinical note + labs + structured EHR risk prediction

Project Question

Can multimodal healthcare AI combine medical images, clinical notes, labs, and structured EHR data to improve risk prediction and clinical interpretability?

Overview Metrics Fairness & Uncertainty **Clinical Interpretation**

Clinical Interpretation

Decision Support, Not Diagnosis

This project is a clinical AI decision-support demonstration. It should not be used for real patient care.

Why Multimodal AI Matters

Clinicians rarely make decisions from one data type alone. They often combine:

- imaging findings
- clinical notes
- lab values
- vitals
- demographics
- prior conditions

A multimodal AI system can model these combined signals.

What Metadata and Labs Improve

Structured EHR features and lab values may capture clinical context that is not visible in medical images, such as inflammation, comorbidity burden, abnormal vitals, or prior conditions.

Failure Modes

The model may fail when:

- training data is biased
- notes contain ambiguous language
- imaging quality differs
- labs are missing
- patient subgroups are underrepresented
- labels are noisy
- synthetic demo data does not reflect real clinical complexity

Fairness and Bias

Fairness should be evaluated across subgroups such as age band, sex, site, race/ethnicity when ethically and legally appropriate, and insurance or socioeconomic proxies only with careful governance.

Human Review

A real deployment would require clinician validation, external validation, calibration, fairness review, privacy review, and regulatory review.